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# Yeshan Qiu

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## EDUCATION

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**National University of Singapore (NUS)**, Singapore, Singapore 01/2022-Present

*PhD Candidate in Architecture, expected 2026*

**Shanghai Jiao Tong University (SJTU)**, Shanghai, China 09/2018-03/2021

*Master's Degree in Landscape Architecture*

*Dissertation: Construction and Design Application of Bayesian Model of Public Ecological Aesthetic Perception in Urban Green Space Based on Deep Learning*

**Hainan University (HNU)**, Haikou, China 09/2014-06/2018

*Bachelor's Degree in Landscape Architecture*

Overall GPA: 3.82/4.0

Ranking: 1/136

## RESEARCH INTERESTS

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- Urban biodiversity, avian ecology, and biodiversity modelling
- Urban landscape patterns, 3D vegetation structure, and pattern–process relationships
- Ecological aesthetics, human perception, and ecosystem service assessment

## PUBLICATIONS

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### Manuscripts under review / in preparation for submission

**Qiu, Y.**, Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Systematic Review of the Relationships between Multidimensional Urban Patterns and Avian Biodiversity: A Meta-analysis [Manuscript under review].

**Qiu, Y.**, Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Mapping Methods Matter: Influence of Spatial Analytical Units on Urban Avian Biodiversity Assessment: A Meta-analysis [Manuscript under review].

**Qiu, Y.**, Chi, D., Hahs, A. K., Richards, D. R., Yang, J., Meili, N., Zhu, Y., Wang, J., Lin, B. B., Manoli, G., Zhang, Y., Burlando, P., McDonnell, M. J., & Tan, P. Y. (2026). Assessing Urban Pattern Effects on Bird Species Richness Across 68 Cities [Manuscript in preparation for submission].

### Peer-reviewed journal articles

**Qiu, Y.**, Chi, D., Meili, N., Zhu, Y., Wang, J., McDonnell, M., & Tan, P. Y. (2024). Exploring the Impact of Urban Pattern on Avian Occurrence: A Buffer-Based Analysis Using Citizen Science Data. AGU Fall Meeting Abstracts, 153, B43B-153.

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**Qiu, Y.,** Pan, H., Kalantari, Z., Giusti, M., & Che, S. (2023). The natural focus: Combining deep learning and eye-tracking to understand public perceptions of urban ecosystem aesthetics. *Ecological Indicators*, 156, 111181.

**Qiu, Y.,** Che, S. (2022). Construction Strategies for Urban Ecological Green Space Based on Ecological Aesthetics. *Landscape Architecture*, 29(7), 37-43.

**Qiu, Y.,** Xie, C., Pan, H., & Che, S. (2021). Study of Urban Street-Scene and Landscape Aesthetic Public Perception Based on Deep Learning: The Case of Shanghai. *Chinese Landscape Architecture*, 37(6), 77-81.

**Qiu, Y.,** Chen, Y., & Che, S. (2020). Using Eye-tracking and Deep Learning Approach to Promote Naturalness in Urban Landscape. *EGU General Assembly Conference Abstracts*, EGU2020-6515.

Chi, D., Manoli, G., Lin, B., Aerts, R., Yang, J., Hahs, A., Richards, D., Meili, N., **Qiu, Y.,** et al. (2025). Residential tree canopy configuration and mortality in 6 million Swiss adults: a longitudinal study. *The Lancet Planetary Health*, 9(3), e186-e195.

Meili, N., Zheng, X., Takane, Y., Nakajima, K., Yamaguchi, K., Chi, D., Zhu, Y., **Qiu, Y.,** et al. (2025). Modeling the effect of trees on energy demand for indoor cooling and dehumidification across cities and climates. *Journal of Advances in Modeling Earth Systems*, 17(3), e2024MS004590.

Zhu, Y., Wang, J., Manoli, G., Zhang, Y., Chi, D., Meili, N., **Qiu, Y.,** Lin, G. S., Tan, P. Y., Burlando, P., & Fatichi, S. (2025). Influence of urban form and function on daytime-nighttime population differences and hazard risk assessments. *npj Urban Sustainability*, 5, 91. <https://doi.org/10.1038/s42949-025-00282-0>

Chi, D., Meili, N., Zhu, Y., Wang, J., **Qiu, Y.,** Li, C., Burlando, P., Fatichi, S., Lin, B., et al. (2024). Associations between greenspace patterns and mortality in six million Swiss adults: a longitudinal study. *ISEE Conference Abstracts*, 1.

Meili, N., Zheng, X., Chi, D., Zhu, Y., **Qiu, Y.,** Wang, J., Manoli, G., Burlando, P., et al. (2023). Quantifying Vegetation Effects on Climate and Building Energy Demand in Cities. *AGU Fall Meeting Abstracts*, GC44F-01.

Sun, W., **Qiu, Y.,** Yu, B., & Wang, Y. (2024). Inclusive Design of Smart Landscape in Community Parks for The Migrant Elderly. *Human Factors in Design, Engineering, and Computing*, 159(159).

Chen, Y., **Qiu, Y.,** & Che, S. (2020). Combining deep learning and multi-source data to promote subjective perception of ecosystem services in urban landscape. *EGU General Assembly Conference Abstracts*, EGU2020-6516.

**Qiu, Y.,** Xu, X. (2020). Landscape Design for Modern Healing Spaces under the Influence of Traditional Chinese Health Culture. *Chinese Horticulture Abstracts*, 33(7), 106-108

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## CONFERENCE PRESENTATIONS

**Qiu, Y.,** Chi, D., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2025). Thinking Vertically: How 3D Urban Form Shapes Global Avian Biodiversity: Multi-City evidence for functional guild-specific responses to 3D urban form. *British Ecological Society Annual Meeting 2025*. Poster presentation.

**Qiu, Y.,** Chi, D., Meili, N., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2024). Exploring the Impact of Urban Pattern on Avian Occurrence: A Buffer-Based Analysis Using Citizen Science Data. *AGU Fall Meeting Abstracts*, 153, B43B-153. Poster presentation / Conference abstract.

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**Qiu, Y.,** Chi, D., Meili, N., Zhu, Y., Wang, J., McDonnell, M. J., & Tan, P. Y. (2024). Exploring the Impact of Urban Pattern Metrics on Avian Occurrence: A Buffer-Based Analysis Using Citizen Science Data. Ecological Society of America Annual Meeting2024. Poster presentation.

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## RESEARCH EXPERIENCE

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### **Comparative Ecology of Cities: Understanding Patterns, Processes and Properties of Cities to Improve Urban Conditions**

03/2023-Present

Future Cities Lab Global, NUS, ETH Zurich

*PhD Researcher*

Project Lead: Prof. Dr. Tan Puay Yok, Prof. Dr. Paolo Burlando

- Contributed to a collaborative, multi-module project examining urban pattern–process–function relationships across cities, linking settlement patterns with biodiversity, climate and energy, hydrology, public health, and socio-economic outcomes.
- Independently led the biodiversity module through a cross-city comparative study of 70+ global cities, integrating citizen-science species occurrence records, biodiversity observation data, and spatial pattern metrics into modelling-ready datasets to assess relationships between urban spatial patterns and biodiversity responses.
- Conducted multi-source, multi-city urban pattern metrics analysis, integrating land-use/land-cover data, urban spatial layers, landscape metrics, and 3D urban/vegetation structure indicators.
- Analysed how landscape distribution, composition, configuration, and three-dimensional structure shape biodiversity responses, with implications for planning practices, ecological networks, and nature-based solutions.

### **Collaborative Urban Bird Modelling Projects with NParks**

07/2024- Present

Wildlife Management Research, National Parks Board, Singapore

*Researcher*

Collaborator: Dr. Malcolm Soh, Principal Researcher (Wildlife Management Research)

- Collaborated with NParks' Wildlife Management Research team on applied avian biodiversity modelling projects, including the Decadal Study of Birds in Singapore and straw-headed bulbul population modelling.
- Integrated bird survey data with Landsat-derived land-cover layers, LiDAR-derived vegetation structure metrics, ETH green canopy data, and multi-scale landscape metrics.
- Generated modelling-ready spatial predictors from transect buffers, survey points, and prediction grids, including landscape composition, configuration, canopy connectivity, patch structure, total edge, core area, and percentage cover metrics.
- Supported long-term urban bird community analysis and straw-headed bulbul abundance/population modelling for management-oriented biodiversity assessment and urban wildlife planning.

### **Public Risk Perception, Behaviour Patterns and Public Emotional Guidance under Major Public Health Emergencies**

03/2020-2023

National Natural Science Youth Fund Project

Project No.: 7204100421

*Researcher*

Advisor: Associate Professor Bingqin Yu

- Analysed the impact and potential role of urban green space on public health events and public health;

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- Concluded that the group pressure and psychological restoration of the elderly in the community are strongly correlated with the rate of green space and green vision.

**Synergistic Research on the Application of Street Species in Shanghai and the Compound Benefits of High-Quality Road Greening** 11/2019-09/2020

Shanghai Science and Technology Commission

Project No.: 19DZ1203702

*Team Leader*

Advisor: Professor Shengquan Che

- Responsible for the evaluation of benefits of urban street greening in Shanghai using AI deep learning skills;
- Assisted in optimization of ecological design, including increasing the plant landscape (after evaluation) by 23.2%, seasonal colour richness of plant landscape by 33.67%, and overall green space utilization by 8%;
- Led 6 team members to implement preliminary investigations and assessments of ecologic road environments;
- Designed a natural plant community pattern in urban road green space across the Jing'an district of Shanghai. Independently produced a research report;
- Innovatively utilised ecological natural plant community's mode to replace traditional geometric and regular plant communities in road green spaces to improve the ecological stability of green spaces and increase biodiversity.

**Survey of Shanghai Urban Landscape and Style: Based on Artificial Intelligence and Big Data** 03/2019-11/2020

National Natural Science Youth Fund Project

Project No.: 52000130

*Researcher Assistant*

Advisor: Dr Haozhi Pan

- Interviewed 86 households face-to-face to investigate public perception of urban landscape aesthetics. Used deep learning, streets data and eye tracking technology;
- Analysed street view data with Python to explore 238 road landscapes in downtown Shanghai;
- Set up a street view sample database to conduct eye-movement experiments on 57 local residents in Shanghai, and tracked their saccades, gazes, glances and other physiological index individually;
- Built aesthetic cognition Bayesian model using deep learning to expand the limited aesthetic survey samples with the aim of constructing a rapid and large-scale general survey of public aesthetic preferences;
- Published the survey to core journal papers *Chinese Landscape Architecture* of Beijing University.

**PROFESSIONAL EXPERIENCE**

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**Dawson Landscape Design Co., Ltd.**

01/2018-08/2018

*Landscape Designer Assistant, Landscape design*

- Participated in preliminary analysis of status quo in beautiful village revitalization construction planning and design projects of Haikou city;
- Independently designed Ding'an Community Park planning projects within two weeks and implemented the first draft and final version of the floor plan, which was successfully adopted by clients;

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- Collected data on the site's human history, topography, precipitation, sunlight, and animal and plant biodiversity in the design project of Magnolia Tianxiang Farm. Produced relevant floor plans;
  - Performed modelling analysis by combining GIS and Rhino;
  - Simulated the distribution of light, wind direction and wind speed in four seasons before selecting the best habitat for plants and crops with different living habits;
  - Took full responsibility for the design of the municipal square landscape of Ding'an City, such as site investigation and analysis, floor plan design and design project plan text writing;

## SELECTED HONOURS & AWARDS

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- **First prize** in Academic Scholarship, SJTU 10/2020
- **Second prize** in Academic Scholarship, SJTU 10/2019
- **Third prize** in the National University Landscape Garden Design Competition, CHSLA 10/2019
- Scholarship of National Academic (awarded to **Top 1%**) 11/2017
- **Third prize** in the National University Student Environmental Ecological Science and Technology Innovation Competition (awarded to **Top 3%**) 09/2017
- Outstanding Graduate Student, SJTU 03/2021
- Outstanding Graduate Student, HNU 2015-2017

## ACADEMIC SKILLS

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- **Ecological and biodiversity modelling:** biodiversity modelling, ecological response modelling, avian occurrence analysis, and ecosystem service assessment.
- **Geospatial analysis:** GIS, raster/vector processing, land-use/land-cover analysis, landscape metrics, LiDAR-derived vegetation metrics, and multi-source geospatial data integration.
- **Programming and statistical tools:** R, Python, MATLAB, Netica; statistical modelling, Bayesian modelling, reproducible workflows, and data visualisation.
- **Design and experimental tools:** Adobe Photoshop, Adobe Illustrator, SketchUp, Lumion, WUFI, Tobii eye-tracking equipment, and physiological-response data collection.

## EXTRACURRICULAR ACTIVITIES

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- *Teaching Assistant* for Academic Writing, Integrity and Ethics and Landscape Garden Graduation Design 09/2018-08/2020
- *Leader* of the Bulgarian Rural Research and Planning Practice Project of SJTU's Global Challenge Program 05/2019-08/2019